

Learning C++ from the inventor of C++ – Exercise 1 (Tool Chain)

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Solution

I decided to use both `println` and explicit object parameters for recursive lambda functions in my solution.

This can be seen with this `auto fib` inside the lambda signature `[&](this auto fib, ...)`

To showcase this I created a small Fibonacci sequence that takes a number, `n`, and prints the `n` Fibonacci numbers by recursively calling itself with `fib(num - 1, ...)` inside the lambda functions. Before C++23 this was a lot more cumbersome as lambda functions didn't have a direct name for themselves inside their own body.

Problem

It was only smaller problems I encountered, such as I couldn't understand why `std::println` didn't work at first when I wrote `std::println << "Hello World"` like it works with `cout`. This was due to the oversight that `cout` is an output stream, whereas `println` is a function. Therefore I fixed it by using parenthesis and omitting the unnecessary `std::`. I had also forgotten to state `#include <print>`

For my recursive Fibonacci it was smaller logical mistake I made to get the Fibonacci to work as intended (like printing current instead of `prev1`, and forgetting to print the first 0).